

NETWORK PROGRAMMING LABORATORY

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The aim of this exam is to build a small network diagnostic tool, similar in spirit to simplified versions of tools such as `ping` or `traceroute`. Instead of sending ICMP Echo Requests, the program sends UDP datagrams to a remote IPv4 host and observes the ICMP errors generated in response.

The program must send N UDP datagrams to UDP ports that are expected to be closed. Therefore, the remote host should reply with ICMP *Destination Unreachable, Port Unreachable* messages.

The program should use packet capture through `pcap` to observe both:

- the outgoing UDP datagrams;
- the incoming ICMP Port Unreachable messages.

For each received ICMP error, the program must identify which UDP probe caused it.

Command Line

The program must be invoked as follows:

```
sudo probe-rs <destination_ip> -n <N> -i <interface>
```

Example:

```
sudo probe-rs 192.0.2.10 -n 10 -i eth0
```

This means that the program sends 10 UDP datagrams to host 192.0.2.10, while capturing packets on interface `eth0`.

Sending UDP Probes

The program must send exactly N UDP datagrams. The payload may contain arbitrary data. The program must record the sending time of each probe.

A key point of the exercise is to choose a reliable method for distinguishing the probes when the ICMP errors are received. The chosen method must be explained briefly in comments or in a short note accompanying the solution.

Packet Capture

The program must open a packet capture on the selected interface using the `pcap` crate. A suitable BPF filter is:

```
(dst host <dest_ip> and udp and dst portrange <xxxxx>-<yyyyy>) or (icmp and src host <dest_ip>)
```

The program must capture packets for a finite amount of time, for example a few seconds after the last UDP datagram is sent.

0	15	31
Source Port		Destination Port
Length		Checksum

UDP header

Figure 1: UDP header format.

UDP Header

Before analyzing the ICMP error, it is useful to recall the format of the UDP header. The UDP header has a fixed length of 8 bytes, as shown in Figure 1.

ICMP Header

The program must recognize ICMP Destination Unreachable messages. For IPv4, an ICMP Port Unreachable message has:

`type = 3,      code = 3`

When such a message is received, the ICMP payload contains the complete original IPv4 header and at least the first 8 bytes of the original payload. In this exercise, the original payload is a UDP datagram, so these bytes are sufficient to recover the original UDP header shown in Figure 1.

The basic ICMP header format is shown in Figure 2.

0		31
Type	Code	Checksum
Unused		
Complete original IPv4 header		
First (at least 8) bytes of original payload		

ICMP Destination Unreachable message

Figure 2: ICMP Destination Unreachable header and embedded original packet data.

Packet Processing

The program must process captured outgoing UDP packets with the following structure:

```
Ethernet
  IPv4
    UDP
```

and incoming ICMP errors with the following structure:

```
Ethernet
  IPv4
    ICMP
      embedded original IPv4
      embedded portion of the original UDP
```

For each ICMP Port Unreachable packet, the program must extract enough information from the embedded original packet to decide which probe caused the error.

Packet Field Hints

- For Ethernet:
 - Ethernet header length is 14 bytes;
 - EtherType 0x0800 indicates IPv4.
- For IPv4:
 - the Internet Header Length (IHL) is stored in the low 4 bits of the first byte;
 - IPv4 header length is $IHL * 4$;
 - protocol number 17 indicates UDP;
 - protocol number 1 indicates ICMP.
- For UDP:
 - source port is at offset 0 of the UDP header;
 - destination port is at offset 2 of the UDP header;
 - ports are encoded in network byte order.
- For ICMP:
 - ICMP type is at offset 0;
 - ICMP code is at offset 1;
 - for Port Unreachable, type must be 3 and code must be 3;
 - the embedded original IPv4 packet starts after the first 8 bytes of the ICMP message.

Output

At the end, the program should print a report similar to:

```
UDP probing report
-----
Destination: 192.0.2.10
Probes sent: 10
UDP packets captured: 10
ICMP port unreachable received: 9
Lost replies: 1

Per-probe results:
ID      RTT (ms)
0       2.41
1       2.38
2       2.43
```